Data methodology

How the tables were understood, validated, cleaned and joined · Companion to the Analysis

We got three raw extracts and no data dictionary. The tables join cleanly on ids (99%+ match rates), but they needed 14 fixes before we could trust them: duplicates, broken keys, amounts recorded 1,000× too high, a sign bug in days to close, out-of-range dates and messy labels. Under 1% of rows needed a fix (not counting URL normalization, which applies to every visit). Every rule is in SQL and every count on this page is queried from the data.

1. What each table is

Table	One row is	Key	Rows raw → clean	Date range
sessions	A website visit	`session_id`	307,715 → 307,715	Jan 1 – Jun 30, 2026
signups	A lead signing up for one product	`lead_id`	7,408 → 7,338	Jan 1 – Jul 8, 2026*
opportunities	A closed deal (paid plan)	`opp_id`	1,513 → 1,495	Jan 2 – Jul 26, 2026

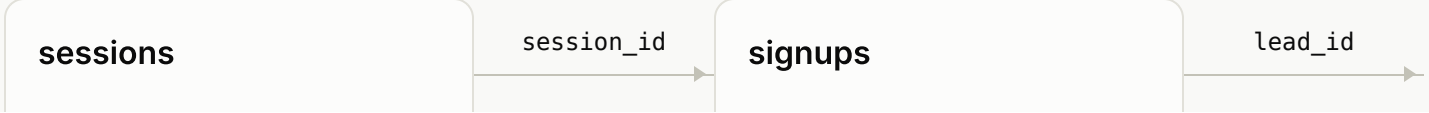
* 8 signups are dated in July, after sessions end on Jun 30. Corrected in R7.

There's no dictionary, so this is our read of each field from the values, and what we checked to confirm it:

Table	Field	What it means (inferred)
`sessions`	`session_id`	Unique visit id. Joins to signups.session_id.
`sessions`	`visitor_id`	Visitor (cookie-level) id; 190,631 distinct, 300 null. A visitor can have many sessions.

sessions	event_dt	Visit date.
sessions	mktg_channel	Marketing channel of the visit. 12 raw labels that are really 7 channels (casing / trailing spaces).
sessions	geo, device_type	Country bucket and device of the visit.
sessions	entry_url	Landing page. 135 raw URLs that are really ~20 pages (case, trailing slash, query string, geo prefix).
sessions	converted_flag	1 if the visit produced a signup. 7,338 sessions flagged; every matched signup's session has flag = 1.
signups	lead_id	Unique lead id. Joins to opportunities.lead_id.
signups	visitor_id, session_id	The visitor and the session the signup happened in (last touch).
signups	created_dt	Signup date. Same day as its session in 100% of matched rows after cleaning.
signups	mktg_channel, geo	Copied from the signup session (agree with sessions in ~100% of rows).
signups	product_code	Product signed up for: STR Starter trial, PRO Pro trial, FCRM Free CRM (from Apr 1).
signups	company_size	Employee count, self-reported. Has sentinels (−5, 999,999).
opportunities	opp_id, lead_id	Closed deal and the lead it came from; at most 1 opportunity per lead.
opportunities	close_dt	Date the deal closed (customer started paying).
opportunities	plan, amount	Plan purchased and monthly price: Starter $35, Pro $110 (stable all period). Some $0 rows.
opportunities	days_to_close	Days from signup to close. Range 1–29 after fixing a sign bug.

2. How the tables connect



307,715 website visits

1 : 0..1

7,338 leads, one product each

1 : 0..1

opportunities
1,495 closed deals

The first instinct with three separate extracts is to line them up by date. We don't need to: the ids connect them row by row, which lets us follow each signup from the visit that produced it to the deal it became. That's what makes the plan and origin cuts in the analysis possible.

Check	Result	Read
Signups that find their session	7,313 / 7,338 (99.7%)	25 have malformed session ids (R8)
Matched signup sessions with converted_flag = 1	7,313 / 7,313 (100%)	Confirms what converted_flag means
Flagged sessions with no signup	25	Same count as the malformed signup keys: the other side of the same break
Signup on the same day as its session	7,313 / 7,313 (100%)	After R7. No attribution window needed
visitor_id agrees between tables	7,310 / 7,313 (99.96%)	3 mismatches; not used for joins
Channel agrees between tables	7,313 / 7,313 (100.0%)	After R1
Geo agrees between tables	7,265 / 7,265 (100.0%)	Where both sides are known (R14)
Opportunities that find their lead	1,480 / 1,495 (99.0%)	15 have malformed lead ids (R13)
Opportunities per lead	max 1	1:1, no fan-out on join
Deals closed before their signup	0	After R7 and R12

Signup product vs plan bought

`product_code` (what the lead signed up for) and `plan` (what they bought) are different fields. Free CRM is not a plan you buy; its upgrades show up as Starter deals.

Signed up for	Bought	Deals
FCRM	Starter	134
PRO	Pro	361
STR	Starter	985

Plan	Amount (after R10)	Deals
Pro	$0.0	8
Pro	$110.0	357
Starter	$0.0	22
Starter	$35.0	1,108

Counts include $0 deals. Prices didn't change during the period, so this is a packaging change, not a pricing change.

3. Cleaning rules

Each rule is numbered in 01_clean.sql with the same id. Rows are flagged rather than dropped when they're still useful for some cuts.

Rule	Table	Issue found	Treatment	
R1	sessions	Channel labels in 12 variants (`'Display '`, `'direct'`, `'Organic search'`)	Trim + map to the 7 canonical channels	1,28
R2	sessions	entry_url has 135 raw values: mixed case, trailing slashes, `?gclid` / `utm` params, geo prefixes (`/uk/`, `/in/`)	Normalize URL, then map to 7 page types (content, home, pricing, product, Free CRM LP, trial LP, campaign LP)	135 raw
R3	sessions	visitor_id missing	Keep (session metrics are fine); flag	3
R4	signups	Exact duplicate rows	Drop	
R5	signups	company_size sentinels: −5 and 999,999	Set to NULL; analyze in buckets, never averages	
R6	signups	Same lead_id twice with conflicting geo / company_size	Keep the most complete row	4 le
R7	signups	created_dt in July, outside the window. Each one's session is in the window, and	Replace with the session date	8

		every other signup happens the same day as its session	
R8	signups	session_id doesn't exist in sessions (format `S9xxxxxxx`)	Keep for signup → paid metrics; excluded only from origin cuts
R9	opportunities	Exact duplicate rows	Drop
R10	opportunities	Amount 1,000× list price (35,000 / 110,000)	Divide by 1,000 (unit error)
R11	opportunities	Amount = $0 (closed but not paying: coupon, comp or internal)	Exclude from paying customers; sensitivity shown with them included
R12	opportunities	Negative days_to_close; the absolute value equals the true date gap	Recompute from dates
R13	opportunities	lead_id doesn't exist in signups (format `G9xxxxxx`). Implied signup dates fall inside the window, so these are broken keys, not pre-January leads	Keep in the finance (close date) view; excluded from cohort conversion
R14	signups	geo holds the literal text `'N/A'` (in addition to true NULLs)	Set to NULL

Checking R7: were the July dates really wrong?

Two of the eight leads with July signup dates have a closed deal. In both, the raw `days_to_close` only makes sense if the signup happened on the session date, not in July. One of them closed in March, which would be before its own signup.

Lead	Raw created_dt	Corrected (session date)	Deal closed	days_to_close (raw)
`G0001922`	Jul 6	Feb 23	Mar 8	13
`G0006974`	Jul 4	Jun 23	Jul 5	12

R2: landing pages after normalizing

Page type	Sessions
content	88,635

home	63,353
product	48,281
pricing	40,767
free_crm_lp	33,349
campaign_lp	22,309
trial_lp	11,021

4. Definitions

- **Paying customer (win):** an opportunity with amount > $0, closed within 30 days of the signup. We call it "paying customer" because finance defines a customer as a closed opportunity, and $0 deals don't pay.
- **Cohort:** signups grouped by the month they signed up. Conversion is always cohort-based (signup month → did they pay), not close-month based.
- **Periods:** Q1 = Jan–Mar 2026 (before Free CRM), Q2 = Apr–Jun 2026 (after launch). Compared as monthly averages.
- **Plan:** the product the lead signed up for (`product_code`), not the plan they bought.
- **Origin attributes** (channel, entry page, device): from the signup session, i.e. last touch. 37% of signups had earlier visits; we don't attribute to those.
- **New vs returning:** returning = the visitor had a session on an earlier day than the signup.

5. Assumptions and why they're safe

- **30-day window.** No deal in the extract closes more than 29 days after signup, so a 30-day window loses nothing. Opportunities run to Jul 26; the 166 signups after Jun 26 have a slightly shorter window. With a 29-day maximum, the impact is a handful of deals in late June.
- **$0 deals excluded.** 30 of 1,495 deals, spread evenly across months. Including them doesn't change any trend (see the finance view in the analysis).

- **Orphan rows kept, not invented.** We don't guess the true keys of the 25 signups and 15 deals with broken ids. They stay in the totals they can support and out of the cuts they can't.
- **January in the finance view.** Some January closes come from December signups, which aren't in the data, so January close-month counts are low. Cohort views aren't affected.
- **Free CRM in the decomposition.** Free CRM has no Q1 conversion rate. The Kitagawa split uses its Q2 rate for both periods, which puts its whole effect in the mix term.

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6. What the data can't tell us

- **Anything inside the product.** Sessions are website visits; there are no product logins or usage events. We can't measure activation.
- **Upgrades after 30 days.** No deal closes more than 29 days after signup, for any SKU, even for April cohorts with ~85 days of history. Either Free CRM users decide in their first month or the extract only captures the first 30 days. A freemium plan usually converts over a longer period, so Free CRM's lifetime conversion could be higher than what we observe.
- **Free CRM landing page before launch.** The Free CRM landing page URLs (`/lp/free-crm`, `/offers/free-crm/`) get traffic from January: 12,802 sessions in Q1 (10% of Q1 traffic), producing 249 Starter and Pro signups before the product existed. The same happens with `/campaign/spring-2026`. It could be a teaser page, a reused URL or an artifact of the extract. Left as is: we can't tell which version is right, and no finding depends on it.
- **Multi-touch attribution.** Only the signup session is linked to the lead.
- **Direct traffic in June.** Direct jumps to 37% of sessions and its signup rate halves. Probably tracking or bots; worth checking at the source.

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7. Reproduce it

The queries are written in Snowflake SQL and avoid engine-specific functions, so they also run unchanged in DuckDB on a laptop. In Snowflake, run 01, 02 and 03 against the `sessions`,

`signups` and `opportunities` tables. Locally, run `00_load_raw.sql` first to load the CSVs into tables with those same names. Run from the folder that contains `data/`.

```
duckdb northwind.duckdb < sql/00_load_raw.sql
duckdb northwind.duckdb < sql/01_clean.sql
duckdb northwind.duckdb < sql/02_funnel.sql
duckdb northwind.duckdb < sql/03_decomposition.sql
```

▶ **00_load_raw.sql — load the CSVs (local only)**

▶ **01_clean.sql — cleaning rules and analysis base table**

▶ **02_funnel.sql — funnel, plan split, slices, finance view**

▶ **03_decomposition.sql — Kitagawa decomposition**